

Selected publications

- [1] Andrew Chu, Xi Jiang, Shinan Liu, Arjun Nitin Bhagoji, Francesco Bronzino, Paul Schmitt, and Nick Feamster. “NetSSM: Multi-Flow and State-Aware Network Trace Generation using State-Space Models”. In: *Proceedings of the ACM on Networking* (2026).
- [2] Augustin Laouar, Paul Lachat, Loïc Desgeorges, Mathieu Cunche, Vincent Roca, Pascale Vicat-Blanc, and Francesco Bronzino. “SoK: Mapping the Privacy Landscape of Geolocation Ecosystems”. In: *Proceedings on Privacy Enhancing Technologies* (2026).
- [3] Taveesh Sharma, Andrew Chu, Paul Schmitt, Francesco Bronzino, Nick Feamster, and Nicole Marwell. “Less is More: Optimizing Probe Selection Using Shared Latency Anomalies”. In: *Proceedings of the ACM on Networking* (2026).
- [4] Xi Jiang, Shinan Liu, Saloua Naama, Francesco Bronzino, Paul Schmitt, and Nick Feamster. “JITI: Dynamic Model Serving for Just-in-Time Traffic Inference”. In: *Proceedings of the ACM on Networking* (2025).
- [5] Taveesh Sharma, Paul Schmitt, Francesco Bronzino, Nick Feamster, and Nicole Marwell. “Beyond Data Points: Regionalizing Crowdsourced Latency Measurements”. In: *Proceedings of the ACM on Measurement and Analysis of Computing Systems* (2025).
- [6] Gerry Wan, Shinan Liu, Francesco Bronzino, Nick Feamster, and Zakir Durumeric. “CATO: End-to-end Optimization of ML Traffic Analysis Pipelines”. In: *USENIX Symposium on Networked Systems Design and Implementation* (2025).
- [7] Youssouph Faye, Francescomaria Faticanti, Shubham Jain, and Francesco Bronzino. “VideoJam: Self-Balancing Architecture for Live Video Analytics”. In: *IEEE/ACM Symposium on Edge Computing* (2024).
- [8] Xi Jiang, Shinan Liu, Aaron Gember-Jacobson, Arjun Nitin Bhagoji, Paul Schmitt, Francesco Bronzino, and Nick Feamster. “NetDiffusion: Network Data Augmentation Through Protocol-Constrained Traffic Generation”. In: *Proceedings of the ACM on Measurement and Analysis of Computing Systems* (2024).
- [9] Manavjeet Singh, Sri Pramodh Rachuri, Bryan Bo Cao, Abhinav Sharma, Venkata Bhumireddy, Francesco Bronzino, Samir R. Das, Anshul Gandhi, and Shubham Jain. “OVIDA: Orchestrator for Video Analytics on Disaggregated Architecture”. In: *IEEE/ACM Symposium on Edge Computing* (2024).
- [10] Shinan Liu, Francesco Bronzino, Paul Schmitt, Arjun Nitin Bhagoji, Nick Feamster, Hector Garcia Crespo, Timothy Coyle, and Brian Ward. “LEAF: Navigating Concept Drift in Cellular Networks”. In: *Proceedings of the ACM on Networking* (2023).
- [11] Francesco Bronzino, Paul Schmitt, Sara Ayoubi, Hyojoon Kim, Renata Teixeira, and Nick Feamster. “Traffic Refinery: Cost-Aware Data Representation for Machine Learning on Network Traffic”. In: *Proceedings of the ACM on Measurement and Analysis of Computing Systems* (2021).
- [12] Francesco Bronzino, Sumit Maheshwari, Ivan Seskar, and Dipankar Raychaudhuri. “NOVN: A named-object based virtual network architecture to support advanced mobile edge computing services”. In: *Elsevier Pervasive and Mobile Computing* (2020).
- [13] Francesco Bronzino, Paul Schmitt, Sara Ayoubi, Guilherme Martins, Renata Teixeira, and Nick Feamster. “Inferring Streaming Video Quality from Encrypted Traffic: Practical Models and Deployment Experience”. In: *Proceedings of the ACM on Measurement and Analysis of Computing Systems* (2019).
- [14] Sumit Maheshwari, Dipankar Raychaudhuri, Ivan Seskar, and Francesco Bronzino. “Scalability and Performance Evaluation of Edge Cloud Systems for Latency Constrained Applications”. In: *IEEE/ACM Symposium on Edge Computing* (2018).
- [15] Kai Su, Francesco Bronzino, KK Ramakrishnan, and Dipankar Raychaudhuri. “MFTP: A Clean-Slate Transport Protocol for the Information Centric MobilityFirst Network”. In: *ACM International Conference on Information-Centric Networking* (2015).

Conferences

- [16] Ayoub Ben Ameer, Francesco Bronzino, Nick Feamster, and Paul Schmitt. “Measuring Low Latency at Scale: A Field Study of L4S in Residential Broadband”. In: *Conference on Passive and Active Network Measurement* (2026).
- [1] Andrew Chu, Xi Jiang, Shinan Liu, Arjun Nitin Bhagoji, Francesco Bronzino, Paul Schmitt, and Nick Feamster. “NetSSM: Multi-Flow and State-Aware Network Trace Generation using State-Space Models”. In: *Proceedings of the ACM on Networking* (2026).
- [17] Loïc Desgeorges, Francesco Bronzino, and Francescomaria Faticanti. “Detection-Aware Controller Placement in Software-Defined Networks”. In: *International Wireless Communications and Mobile Computing Conference* (2026).
- [18] Johann Hugon, Shinan Liu, Paul Schmitt, Nick Feamster, and Francesco Bronzino. “LoFi: Low-Cost Early Application Filter Based on Cached ML Decisions”. In: *IEEE International Conference on Network Softwarization* (2026).
- [2] Augustin Laouar, Paul Lachat, Loïc Desgeorges, Mathieu Cunche, Vincent Roca, Pascale Vicat-Blanc, and Francesco Bronzino. “SoK: Mapping the Privacy Landscape of Geolocation Ecosystems”. In: *Proceedings on Privacy Enhancing Technologies* (2026).
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- [21] Andrea Fox, Francesco De Pellegrini, Francescomaria Faticanti, Eitan Altman, and Francesco Bronzino. “Optimal Flow Admission Control in Edge Computing via Safe Reinforcement Learning”. In: *International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks* (2024).
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- [22] Mathieu Guglielmino, Francesco Bronzino, and Sébastien Monnet. “Outil interactif pour l’alignement de la vue client-opérateur sur les architectures de réseau”. In: *Rencontres Francophones sur la Conception de protocoles, l’évaluation de performances et l’expérimentation de Réseaux de communication* (2023).
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- [23] Martino Trevisan, Idilio Drago, Paul Schmitt, and Francesco Bronzino. “Measuring the Performance of iCloud Private Relay”. In: *Passive and Active Measurement Conference* (2023).
- [24] Ayoub Ben Ameer, Andrea Araldo, and Francesco Bronzino. “On the deployability of augmented reality using embedded edge devices”. In: *IEEE Annual Consumer Communications & Networking Conference* (2021).
- [25] Francesco Bronzino, Nick Feamster, Shinan Liu, James Saxon, and Paul Schmitt. “Mapping the Digital Divide: Before, During, and After COVID-19”. In: *Research Conference on Communication, Information and Internet Policy* (2021).
- [26] Francesco Bronzino, Sumit Maheshwari, Ivan Seskar, and Dipankar Raychaudhuri. “Application-Aware End-to-End Virtualization Using a Named-Object Based Network Architecture”. In: *International Teletraffic Congress* (2021).
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- [28] Francesco Bronzino, Sumit Maheshwari, Ivan Seskar, and Dipankar Raychaudhuri. “NOVN: named-object based virtual network architecture”. In: *International Conference on Distributed Computing and Networking* (2019).
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- [14] Sumit Maheshwari, Dipankar Raychaudhuri, Ivan Seskar, and Francesco Bronzino. “Scalability and Performance Evaluation of Edge Cloud Systems for Latency Constrained Applications”. In: *IEEE/ACM Symposium on Edge Computing* (2018).
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- [32] Shreyasee Mukherjee, Parishad Karimi, Dipankar Raychaudhuri, and Francesco Bronzino. “Enabling Advanced Network Services in the Future Internet Using Named Object Identifiers and Global Name Resolution”. In: *International Conference on Communication Theory, Reliability, and Quality of Service* (2017).
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- [38] Razanne Abu-Aisheh, Francesco Bronzino, Lou Salaün, and Thomas Watteyne. “CARA: Connectivity-Aware Relay Algorithm for Multi-Robot Expeditions”. In: *MDPI Sensors* (2022).
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Workshops

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- [43] Xi Jiang, Shinan Liu, Saloua Naama, Francesco Bronzino, Paul Schmitt, and Nick Feamster. “Towards Designing Robust and Efficient Classifiers for Encrypted Traffic in the Modern Internet”. In: *IAB workshop on Management Techniques in Encrypted Networks* (2022).
- [44] Shinan Liu, Francesco Bronzino, Paul Schmitt, Arjun Nitin Bhagoji, Nick Feamster, Hector Garcia Crespo, Timothy Coyle, and Brian Ward. “Understanding Model Drift in a Large Cellular Network”. In: *Practical Adoption Challenges of ML for Systems in Industry Workshop* (2022).
- [45] Razanne Abu-Aisheh, Francesco Bronzino, Myriana Rifai, Lou Salaün, and Thomas Watteyne. “Coordinating a Swarm of Micro-Robots Under Lossy Communication”. In: *2nd ACM International Workshop on Nanoscale Computing, Communication, and Applications*. 2021.
- [46] Razanne Abu-Aisheh, Myriana Rifai, Francesco Bronzino, and Thomas Watteyne. “(POSTER) Impact of Connectivity Degradation on Networked Robotic Swarm Cooperation”. In: *17th International Conference on Distributed Computing in Sensor Systems*. 2021.
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- [50] Razanne Abu-Aisheh, Francesco Bronzino, Myriana Rifai, Brian Kilberg, Kris Pister, and Thomas Watteyne. “Atlas: Exploration and Mapping with a Sparse Swarm of Networked IoT Robots”. In: *16th International Conference on Distributed Computing in Sensor Systems*. 2020.
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- [77] Razanne Abu-Aisheh, Francesco Bronzino, Myriana Rifai, Brian Kilberg, Kris Pister, and Thomas Watteyne. “Exploration and Mapping using a Sparse Robot Swarm: Simulation Results”. PhD thesis. Inria, 2020.
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Software and datasets

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